

LESSON 5-8 • SESSION 2 • PREREQUISITE PRACTICE B • SKILL 1 OF 2

Surface Area of Pyramids

Builds on Lesson 5-8 Session 1 • 6.G.A.4

TODAY'S LESSON

You will find lateral area: surface area without the base, like the open tent.

YOU NEED TOFind a square pyramid's surface area: base + 4 faces. **You will use it in Practice Form A, problems 3 and 6.****STEPS TO FOLLOW***use these same steps on every problem*

- 1 Base:** side $\times$ side.
- 2 Faces:** $\frac{1}{2} \times$ base edge $\times$ slant height, then $\times 4$.
- 3 Add** the base and the 4 faces.

WATCH ME*the steps in action***A square pyramid has base 11 in and slant height 14 in. Find its surface area.****Step 1** Base: $11 \times 11 = 121$.**Step 2** One face: $\frac{1}{2} \times 11 \times 14 = 77$. Four faces: 308.**Step 3** $121 + 308 = 429$.**Answer 429 in²****TRY IT WITH HELP***fill in each step*

- 1.** A square pyramid has base 6 cm and slant height 7 cm. Find its surface area.

Step 1 Base: $6 \times 6 =$ _____**Step 2** One face: $\frac{1}{2} \times 6 \times 7 =$ _____. Four faces: _____**Step 3** $36 + 84 =$ _____ cm²

- 2.** A square pyramid has base 8 m and slant height 13 m. Find its surface area.

Step 1 Base: $8 \times 8 =$ _____**Step 2** One face: $\frac{1}{2} \times 8 \times 13 =$ _____. Four faces: _____**Step 3** $64 + 208 =$ _____ m²**YOUR TURN***use the steps on your own*

- 3.** Find each surface area (base, slant height).

a. base 5, slant 6 _____**b.** base 10, slant 12 _____**c.** base 4, slant 9 _____**d.** base 12, slant 15 _____

Area of Equal Triangles

Builds on Lesson 5-4 Session 1 • 6.G.A.1

TODAY'S LESSONYou will find lateral area: one slanted face $\times$ the number of faces.**YOU NEED TO**Find the area of equal triangles: one triangle $\times$ how many. **You will use it in Practice Form A, problems 2, 4, and 5.****STEPS TO FOLLOW***use these same steps on every problem*

- 1 Find** one triangle: $\frac{1}{2} \times b \times h$.
- 2 Count** the equal triangles.
- 3 Multiply:** one triangle $\times$ the count.

WATCH ME*the steps in action***3 equal triangles each have $b = 16$ cm and $h = 11$ cm. Find the total area.****Step 1** $\frac{1}{2} \times 16 \times 11 = 88$.**Step 2** Count: 3.**Step 3** $3 \times 88 = 264$.**Answer** **264 cm^2** **TRY IT WITH HELP***fill in each step***4.** 4 equal triangles each have $b = 10$ in and $h = 7$ in. Find the total area.**Step 1** $\frac{1}{2} \times 10 \times 7 = \underline{\hspace{2cm}}$ **Step 2** Count: $\underline{\hspace{2cm}}$ **Step 3** $4 \times 35 = \underline{\hspace{2cm}} \text{ in}^2$ **5.** 3 equal triangles each have $b = 20$ ft and $h = 13$ ft. Find the total area.**Step 1** $\frac{1}{2} \times 20 \times 13 = \underline{\hspace{2cm}}$ **Step 2** Count: $\underline{\hspace{2cm}}$ **Step 3** $3 \times 130 = \underline{\hspace{2cm}} \text{ ft}^2$ **YOUR TURN***use the steps on your own***6.** Find the total area.**a.** 4 triangles, $b = 14$, $h = 9$ $\underline{\hspace{2cm}}$ **b.** 3 triangles, $b = 12$, $h = 25$ $\underline{\hspace{2cm}}$ **c.** 4 triangles, $b = 26$, $h = 15$ $\underline{\hspace{2cm}}$ **d.** 6 triangles, $b = 10$, $h = 8$ $\underline{\hspace{2cm}}$ **READY CHECK***a problem like the one in today's lesson***7.** A square pyramid has base 10 in and slant height 12 in. What is its lateral area?**A** 60 in^2 **B** 240 in^2 **C** 340 in^2 **D** 480 in^2 **CHECK YOURSELF**
☐ I can do this on my own
 ☐ I need a little help
 ☐ I have a question about # $\underline{\hspace{1cm}}$